Charles Nguyen

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EDUCATION

University of California, San Diego — *La Jolla, CA*
Bachelor of Science in Computer Science — *June 2026*

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Rust, Python, SQL
Frontend: React, React Native, Expo, Astro, TanStack Start
Backend & Infra: Convex, Firebase, Docker, Linux, Google APIs
Auth & AI: OIDC, OAuth 2.0, JWT, Logto, OpenRouter, Okta Certified Professional

EXPERIENCE

IEEE Student Branch at UC San Diego — *San Diego, CA*
Webmaster — *Oct. 2023 – May 2026*

- Architected and maintained a centralized member and officer platform using TanStack Start, TypeScript, Convex, and Google APIs, supporting events, reimbursements, sponsorships, analytics, resume access, and user administration.
- Implemented organization-wide authentication using Logto, OpenID Connect, JWTs, and role-based access control, securing workflows and permissions for over 800 members and over 80 officers.

Halıcıoğlu Data Science Institute, UC San Diego — *Hybrid*
HDSI Lab 3.0 Fellow — *Nov. 2025 – June 2026*

- Built and launched the Lab 3.0 website using Astro and Markdown-driven content, delivering responsive pages, dynamic member listings, generated tables of contents, and privacy documentation.
- Built an Astro/Convex code-sharing platform for high-school partnerships with versioning, syntax highlighting, logs, and diffs.

ElitAxis LLC — *Remote*
Contract Developer — *June 2026 – Present*

- Built client websites with Next.js, Astro, and Convex, adding Stripe and Shippo ecommerce and a client CMS.
- Implemented company server authentication, Docker containerization, and Google Workspace administration.

PROJECTS

AI Expense Tracker | *TypeScript, React Native, Expo, Convex, OpenRouter* — *Feb. 2026 – Present*

- Built a receipt-processing app extracting merchants, line items, taxes, tips, and totals with review and friend bill splitting.
- Built OpenRouter vision extraction with schema validation, fallback retries, confidence checks, and 98% accuracy.
- Implemented authenticated multi-user groups, expense reconciliation, and persistent storage with Convex and Better Auth, reducing manual entry time by 3–5 minutes per scanned receipt. Currently in beta on Apple TestFlight and Google Play Testing.

IEEE Presence Tracker | *Rust, Convex, Raspberry Pi, BlueZ, Docker* — *Jan. 2026 – June 2026*

- Built a Raspberry Pi Bluetooth presence system with Rust daemon, Convex backend, and web dashboard for app-free check-ins.
- Implemented l2ping and connect-probe detection with hysteresis/flap freeze, syncing attendance to Convex and Discord/Slack.

LEADERSHIP

IEEE Student Branch at UC San Diego — *San Diego, CA*
Vice Chair of Projects — *Jan. 2026 – May 2026*

- Led robotics, HPC, and engineering programs for hundreds of students, coordinating leads, budgets, inventory, and milestones.
- Secured more than $10,000 for TritonBots and standardized bills of materials, parts allocation, and inventory workflows.